1ST SEMESTER 2024

(Please write your Exam Roll No.) *Exam R. No. .....*

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] JANUARY 2024

Paper Code: ES-101 ***Subject: Programming in 'C'***

Time: 3 Hours ***Maximum Marks:*** 60

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Assume missing data if any.

Q1 Attempt all questions:- **(2x6=12)**

(a) What is recursion? Differentiate direct and indirect recursion?

(b) What is the return value of f(p,p), if the value of p is initialized to 5 before the call? Note that the first parameter is passed by reference, whereas the second parameter is passed by value.

```
int f (int &x, int c)
{
    c = c - 1;
    if (c==0) return 1;
    x = x + 1;
    return f(x,c) * x;
}
```

(c) Discuss the structure of the C program with examples.

(d) What is printed by the print statements in the program P1 assuming call by reference parameter passing?

```
Program P1()
{
  x = 10; y = 3;
  func1(y,x,x);
  print x; print y;
}
func1(x,y,z)
{
  y = y+4;
  z = x+y+z;
}
```

(e) Write a C program to print the Fibonacci series.

(f) Explain file handling. Differentiate sequential and random-access files.

UNIT-I

Q2 (a) What is an operator? Explain different types of operators. Discuss arithmetic, relational, logical, bitwise, assignment and comma operators in detail. **(6)**

(b) Explain storage classes. Elaborate auto, extern, static and register storage classes with the help of examples. **(6)**

Q3 (a) What is a token? Explain different types of tokens. Discuss keywords, identifiers, constants, and special characters in detail with examples. **(3)**

(b) What is a variable? Discuss the scope, visibility and lifetime of a variable. **(3)**

(c) What is the difference between call by value and reference? Write a C program to elaborate on their difference. **(6)**

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UNIT-II

Q4 (a) What are jump statements? Discuss control structure. Explain different types of control structures. Elaborate the difference between while and do while loop control structure with the help of C program. **(6)**

(b) What is a 2D array? Explain the declaration and initialization of a two-dimensional array. Write a C program to initialize and print values using the 2D array. **(6)**

Q5 (a) What is a function? Explain elements of functions. Discuss different types of functions. Write a C program to elaborate on function definition, function call and function prototype. **(6)**

(b) What is a string? How string can be declared and initialized? Discuss the following string handling functions with examples. **(6)**

(i) strlen() (iv) strcpy()
(ii) strrev() (v) strspn()
(iii) strcat() (vi) strpbrk()

UNIT-III

Q6 (a) Explain file handling. What is the need for file handling? Discuss file access modes for text files: r, w, a, r+, w+, a+ and binary files: rb, wb, ab, r+b, w+b, a+b. **(6)**

(b) What is a pointer? How to declare and initialize a pointer? Discuss the significance. **(6)**
of * & operator. Write a C program to add two variables using Pointer.

Q7 (a) Explain structure and union? What is the difference between structure and union? Write a C program to elaborate on their difference. **(6)**

(b) Discuss file handling functions for input-output operations. Explain file positioning random access functions. Differentiate text and binary files. **(6)**

UNIT-IV

Q8 (a) What is a header file? Explain library functions of following header files in detail. **(6)**

(i) stdio.h (iv) ctype.h
(ii) stdlib.h (v) math.h
(iii) conio.h (vi) string.h

(b) What is binary search? Search item 23 from the following sorted data elements using binary search. **(6)**

2, 5, 8, 12, 16, 23, 38, 56, 72, 91

Q9 (a) What is insertion sort? Write a C program to implement insertion sort. **(6)**

(b) Explain selection sort. Sort the following numbers using selection sort. **(6)**

77, 33, 44, 11, 88, 22, 66, 55

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